

Autonomous System (ASN)-----A group of Routers under single Administrative control

IANA Internet Assigned Numbering Authority ---IPs & ASN

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EIGRP VVIMP

1. Enhanced Interior Gateway Routing Protocol

Cisco Prop

2. AD Value 90 & 170

3. Supports VLSM

4. classless RP

5. Cisco Prop RP

6. Best IGP RP because of working features

7. Supports IP subnet Zero

8. Hybrid (Adv. DVRP)

9. Multicast address as 224.0.0.10

10. Triggered Update

11. Incremental Update

12. Metrics: BW, Delay of Line, Load, MTU, Reliability

13. Conditions for neighborship relationship

1. AS number should match

2. Hello and ACK should receive

3. Metric should be same

14. hello packets are sent after every 5 secs (keep alive packets) 224.0.0.10

15. supports auto & manual summarization

Types of EIGRP Tables:

1. Routing table: best path sh ip route eigrp

2. neighbor table: neighbor infn

3. topology table: alternate best path infn is kept in topology table (6
alternate best path is kept)

16. EIGRP uses DUAL

Diffusing update algorithm

17. Types of EIGRP Packets:

hello & ack

route update

query and reply

neighbor discovery /recovery reliable packet

contains routing info

neighbor discovery /recovery query packets--

multicast, reply--unicast

18. EIGRP uses RTP Reliable Transport protocol

19. Hop count: 100 max 255

EIGRP K Metrics:

K Value

Component

Description

K1 Bandwidth

Lowest bandwidth of route

K2 Load

Worst load on route based on packet rate

K3 Delay

Cumulative interface delay of route

K4 Reliability

Worst reliability of route based on keep alive

K5 MTU

Smallest MTU in path [Not used in route

calculation]

#sh ip protocols

K1 (Bandwidth): 1

K2 (Load): 0

K3 (Delay): 1

K4 (Reliability): 0

K5 (MTU): 0

Successor Route: best path to reach destination nw (routing table)

Feasible successor: Alternate best paths to reach destination(topology table)

Feasible Distance: total distance from source to destination

Reported Distance/AD(Advertised Distance): total distance from next hop till destination
nw is called RD or AD (Advertised Distance)

Routing with EIGRP

on First Router:

```
#sh ip route
#conf t
config# router eigrp 1                                # 1--->AS number (any number between 1-65535)
config-router# network 5.0.0.0
config-router# network 6.0.0.0
config-router# network 192.168.1.0
config-router# network 192.168.1.96
config-router# no auto-summary
config-router# end
# sh ip route eigrp
```

on Second Router:

```
#sh ip route
#conf t
config# router eigrp 1
config-router# network 192.168.1.96
config-router# network 192.168.1.32
config-router# network 192.168.1.128
config-router# no auto-summary
config-router# end
# sh ip route eigrp
```

on Third Router:

```
#sh ip route
#conf t
config# router eigrp 1
config-router# network 192.168.1.128
config-router# network 192.168.1.64
config-router# no auto-summary
config-router# end
# sh ip route eigrp
```

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Loopback Interface: is a logical interface

```
conf t
int lo0
ip add 5.0.0.1 255.0.0.0
```

```
exit
int lo1
ip add 9.0.0.1 255.0.0.0
```

```
exit
config#router eigrp 10
                network 5.0.0.0
                network 9.0.0.0
```

```
do sh ip route
```

```
goto pc5                ping 9.0.0.1
                        ping 5.0.0.1
```

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OSPF Open Shortest Path First

1. Example of LSRP(dynamic routing protocol)
2. Classless RP

3. AD value:110
4. manual Summarization
5. Open Standard RP (cisco & non cisco)
6. supports VLSM
7. works on the concepts of area (dividing large internetworks into small internetworks)
8. Hop count= unlimited
9. Dr. Dijkstra Algorithm
10. Supports Ipv4 & IPv6 too
11. supports multi-vendor routers
12. consumes huge amount of CPU and RAM power.
13. Hello packets and ack
14. Multicast address as 224.0.0.5 & 224.0.0.6
15. triggered update (incremental update)
16. Condition for neighborship
 1. Area id should be same
 2. authentication pwd
 3. hello & dead interval should be same
17. Metric: cost (1/BW)
18. LSA-Link State Advertisement is used by OSPF --- LSA are data packets to exchange network topology information with other areas.

The command to display OSPF neighbors on a Cisco device is

```
# show ip ospf neighbor
```

R1	R2
10.0.0.0/8	20.0.0.0/8
172.16.0.0/16	192.168.1.0/24

process id is local to ospf router

```
R1
config # router ospf 1
config-router)# network 10.0.0.0 0.255.255.255 area 0
config-router)# network 172.16.0.0 0.0.255.255 area 0
config-router)# do sh ip route ospf
```

```
R2
config # router ospf 50
config-router)# network 20.0.0.0 0.255.255.255 area 0
config-router)# network 192.168.1.0 0.255.255.255 area 0
config-router)# do sh ip route ospf
```

=====

on First Router:

```
#sh ip route
#conf t
config# router ospf 5
config-router# network 5.0.0.0 0.255.255.255 area 0
config-router# network 6.0.0.0 0.255.255.255 area 0
config-router# network 192.168.1.0 0.0.0.31 area 0
config-router# network 192.168.1.96 0.0.0.3 area 0
config-router# end
# sh ip route ospf
# sh ip ospf int s0/0
```

on Second Router:

```
#sh ip route
#conf t
config# router ospf 10
config-router# network 192.168.1.96 0.0.0.3 area 0
config-router# network 192.168.1.32 0.0.0.31 area 0
```

```
config-router# network 192.168.1.128 0.0.0.3 area 0
config-router# end
# sh ip route ospf
```

on Third Router:

```
#sh ip route
#conf t
config# router ospf 15
config-router# network 192.168.1.128 0.0.0.3 area 0
config-router# network 192.168.1.64 0.0.0.31 area 0
config-router# end
# sh ip route ospf
```

OSPF Hello & Dead interval timers:

Default Timer values

Hello 10 secs

Dead 40 secs

To verify OSPF status(adjacency and timers) of s0/0

```
#sh ip ospf int s0/0
```

To change hello interval of s0/0 of first router:

```
config# int s0/0
      ip ospf hello-interval 30
```

now adjacency will be removed, because hello timers are now different between two OSPF speaking routers(connected point-point)

Again set the timer value back to 10

```
config# int s0/0
      # ip ospf hello-interval 10
```

now adjacency will be formed

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Redistribution between IGPs: Redistribution means routing between different IGPs, Injecting routes of one routing protocol into other

Redistribution between

1. RIP-EIGRP
2. EIGRP-OSPF
3. OSPF- RIP

on Redistribution router

```
router rip
redistribute eigrp 200 metric 3
exit
```

```
router eigrp 200
```

```
redistribute rip metric 1 1 1 1 1
```

OR

```
redistribute rip metric 500 0 255 1 1500
                        bw dly rel load MTU
```

```
do sh ip route
```

